BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY(BIT)

FACULTY OF COMPUTING

Department of Software engineering

Course Title: Operating System and System Programming(Seng2034)

Given Installation Title: Parrot security OS

Individual Assignment

2ND YEAR

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Introduction

In today's fast-evolving technological environment, operating systems serve as the foundation of all computing activities. They are responsible for managing hardware resources, coordinating software execution, and ensuring secure and efficient communication between different components of a computer system.

With the rapid growth of cyber threats such as hacking, malware, and data breaches, the importance of security-focused operating systems has increased significantly. These specialized systems are designed to help professionals identify vulnerabilities, test system defenses, and enhance cybersecurity measures.

One such powerful system is **Parrot Security OS**, which is based on Debian Linux. It is specifically developed for cybersecurity purposes, including penetration testing, ethical hacking, digital forensics, and privacy protection. It includes a wide range of pre-installed tools that make it highly useful for both students and professionals in the field of information security.

The main motivation behind this project is to gain practical, hands-on experience in installing and working with a specialized operating system in a controlled and safe environment. Instead of installing the OS directly on physical hardware, virtualization tools allow us to simulate a real system without risking damage to the host machine.

This project also focuses on understanding low-level system operations, especially networkingrelated system calls like shutdown(), which play an important role in managing communication between systems.

1. Objectives

The main objective of this project is to move beyond theoretical understanding and gain practical experience in working with a real-world operating system designed for cybersecurity. This project is not only about installing an operating system, but also about understanding how it functions internally and how it can be used effectively in a secure computing environment.

To achieve this, the project focuses on several important goals. First, it aims to successfully install and configure Parrot Security OS within a virtual environment using tools such as VirtualBox or VMware. This includes understanding each step of the installation process, from setting up the virtual machine to configuring system resources like memory, storage, and processors.

Another key objective is to develop a deeper understanding of system-level operations. This includes exploring how the operating system manages processes, memory, and communication, with special attention given to the `shutdown()` system call. By studying this function, the project seeks to understand how systems control network communication and ensure proper termination of connections.

The project also aims to examine the different file systems supported by Parrot Security OS and understand their structure, performance, and real-world applications. This helps in recognizing why certain file systems, such as ext4, are commonly preferred in Linux environments.

In addition, an important objective is to identify and overcome challenges encountered during the installation process. By troubleshooting issues such as system errors, performance problems, and configuration difficulties, the project enhances problem-solving skills and builds confidence in handling technical issues.

Furthermore, the project evaluates the strengths and limitations of Parrot Security OS, particularly in the context of cybersecurity tasks. This includes analyzing its usability, performance, and suitability for beginners and advanced users.

Finally, the project aims to build a solid understanding of virtualization technology—how it works, why it is important, and how it allows multiple operating systems to run safely on a single machine. This knowledge is essential for modern computing environments and future professional work in IT and cybersecurity.

Overall, this project is designed to develop both technical knowledge and practical skills, preparing the learner for more advanced studies and real-world applications

2. Requirements

i. Hardware Requirements

To run Parrot Security OS smoothly in a virtual environment, the following hardware configuration is recommended:

- A processor with virtualization support (Intel VT-x or AMD-V)
- Minimum of 4 GB RAM (8 GB recommended for better performance)
- At least 40 GB of free disk space Stable internet connection

ii. Software Requirements

 Virtualization Software:
- Oracle VM VirtualBox or VMware Workstation OS

Image: ○ Parrot Security OS ISO file Host OS:
- Windows / Linux / macOS

3. Installation Steps

Step 1: Download Required Files

First, download Oracle VM VirtualBox and the Parrot Security OS ISO file from their official websites. These two files are required to create and install the virtual machine successfully.

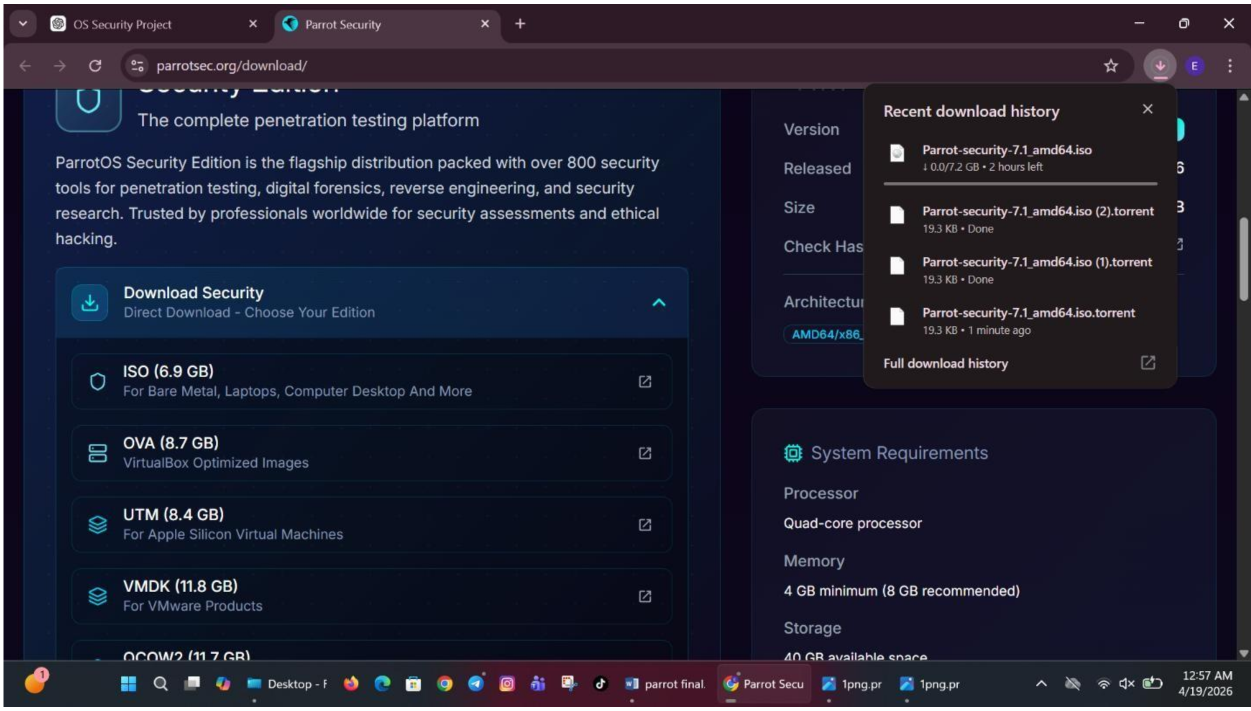

Figure 1: Downloading Parrot Security OS ISO file from official website.

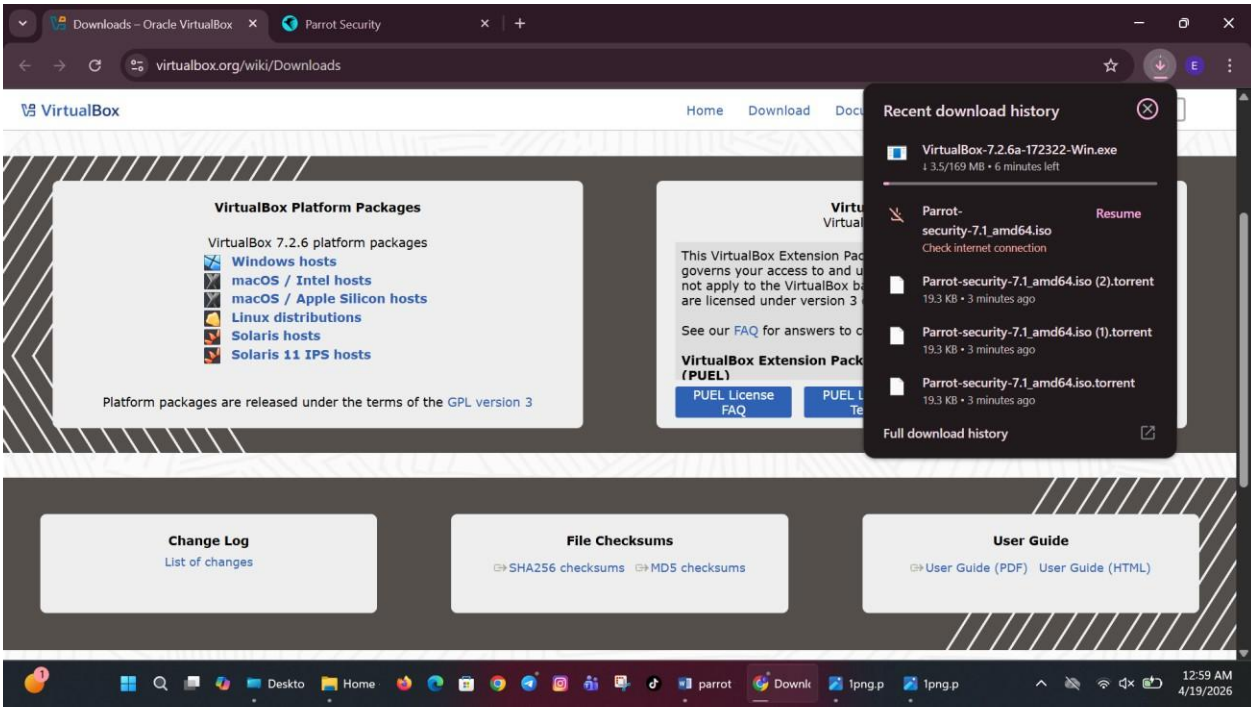

Figure 2: Downloading VirtualBox setup on Windows system.

Step 2: Install VirtualBox

Open the VirtualBox setup file and begin the installation process. Follow the default setup instructions by clicking “Next” until the installation is completed.

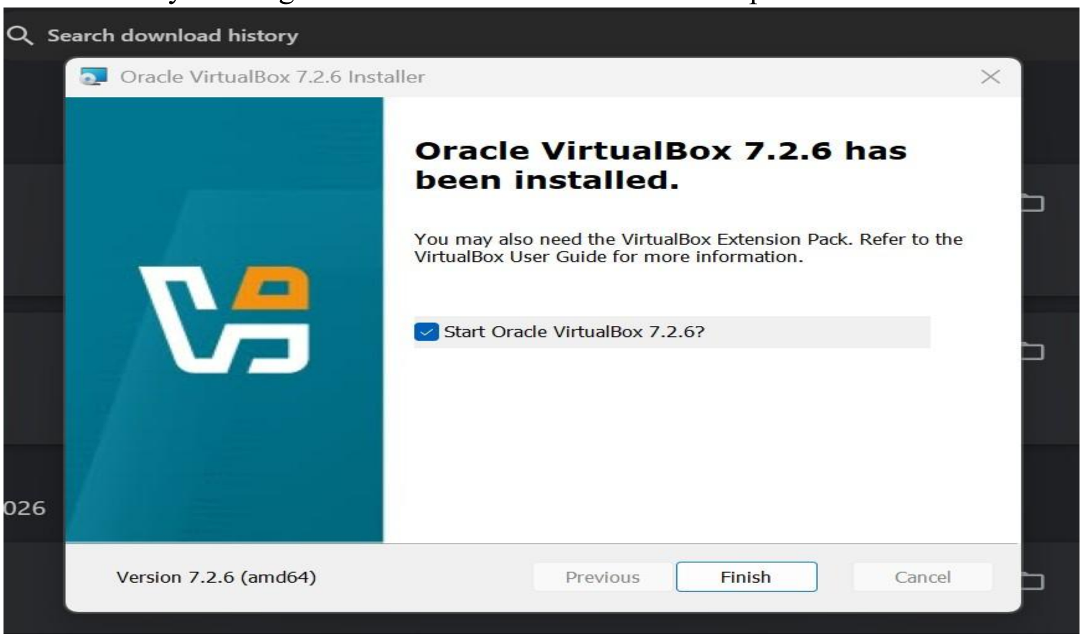

Figure 3: Installing VirtualBox setup on Windows system.

Step 3: Create a New Virtual Machine

Open VirtualBox and click on the “New” button. Enter the virtual machine name as required, such as Parrot Security OS, and select Linux as the operating system type.

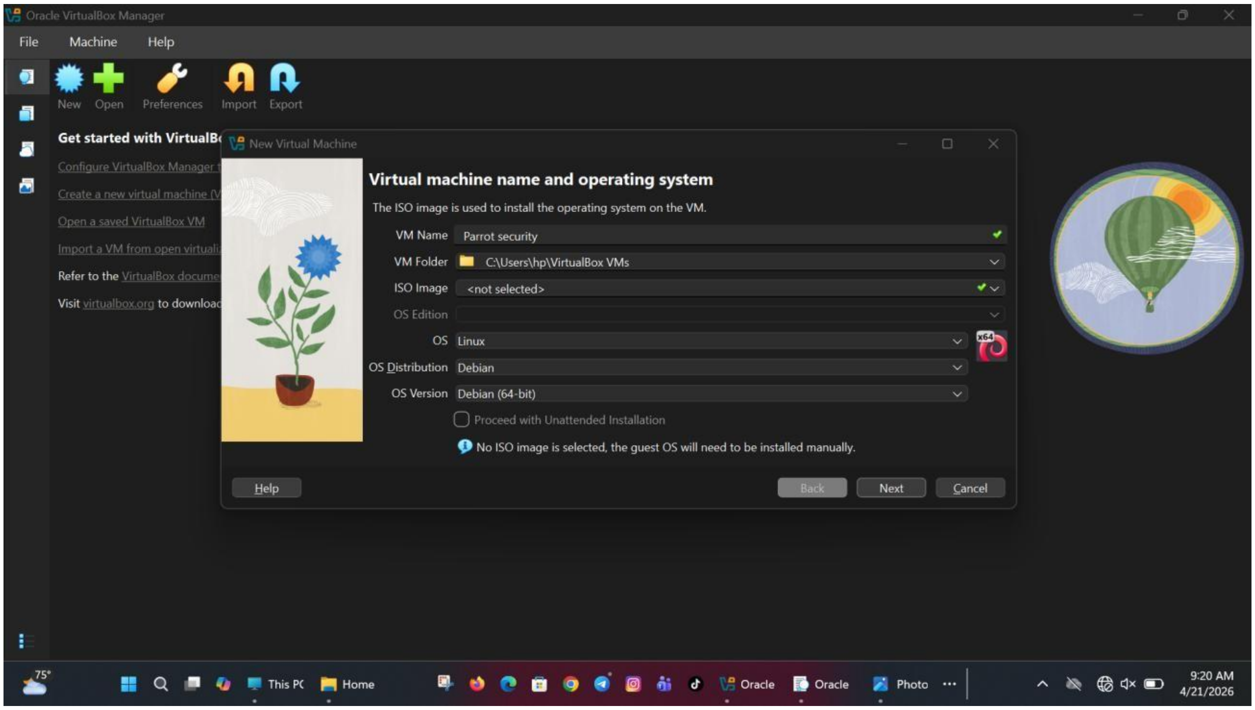

Figure 4: Creating a new virtual machine in VirtualBox.

Step 4: Allocate Memory

Assign RAM memory to the virtual machine based on your system capacity. Giving more RAM helps improve performance and makes the virtual machine run more smoothly.

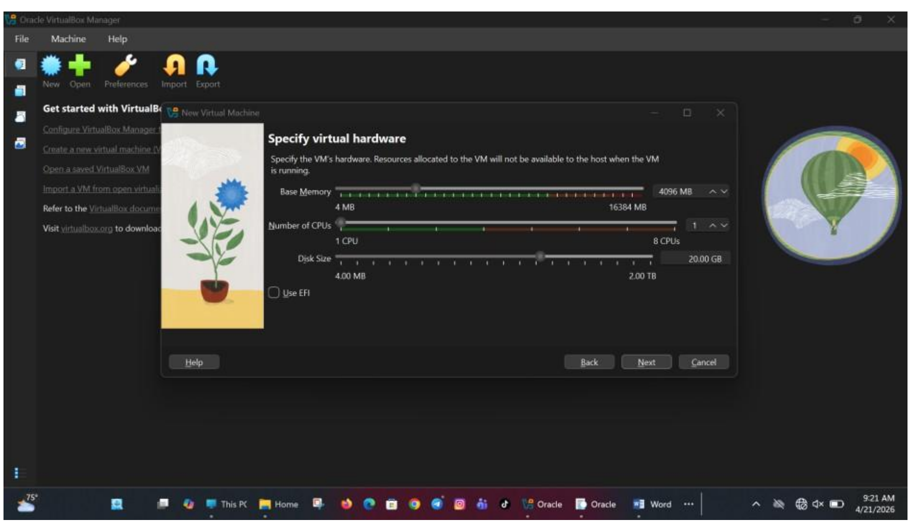

Figure 5: Allocating RAM memory for virtual machine

Step 5: Create Virtual Hard Disk

Create a virtual hard disk where the operating system files will be stored. Choose VDI format and select dynamic allocation to save storage space.

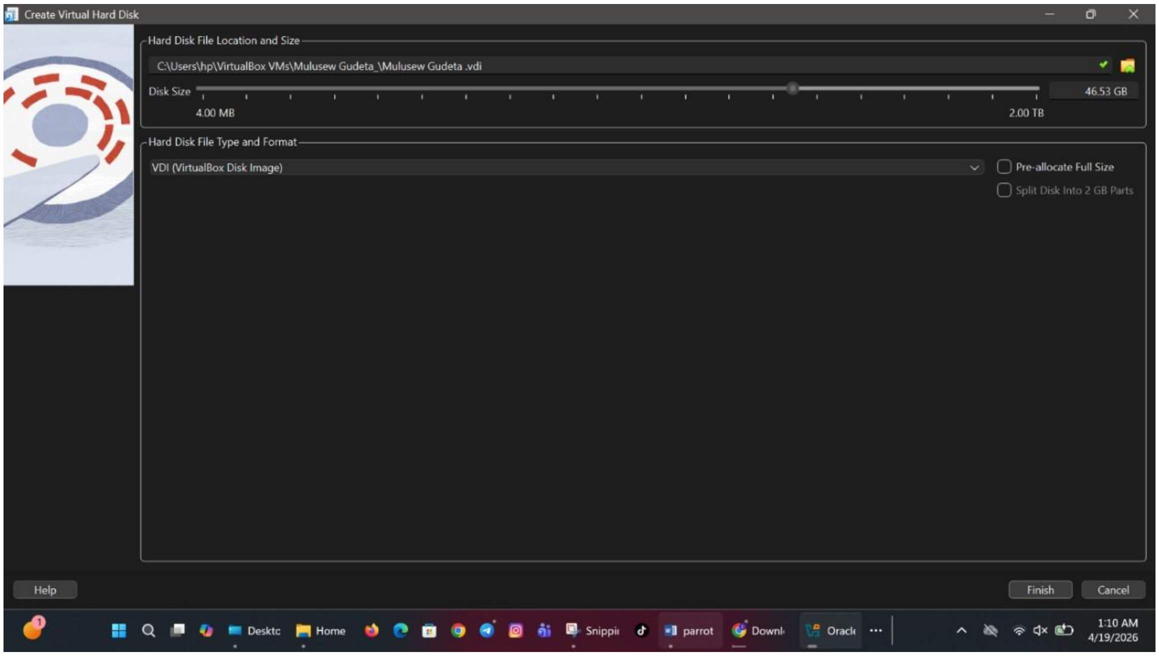

Figure 6: Creating virtual hard disk storage.

Step 6: Attach ISO File

Go to the storage settings of the virtual machine and attach the downloaded Parrot Security OS ISO file. This allows the system to boot and start the installation process.

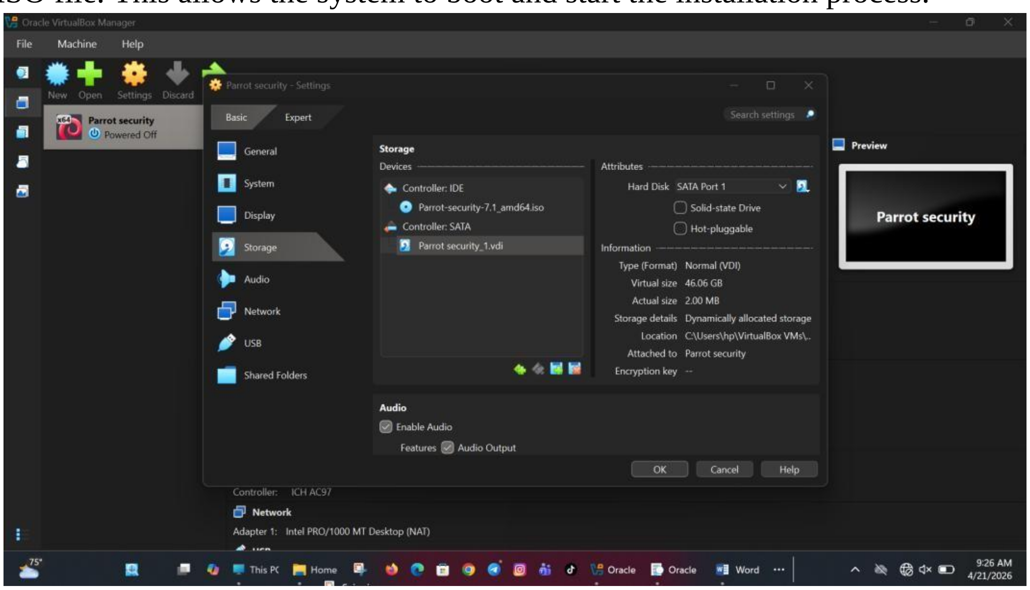

Figure 7: Attaching Parrot OS ISO file.

Step 7: Start the Virtual Machine

Click the “Start” button to run the virtual machine. From the boot menu, select the “Install” option to begin installing the operating system.

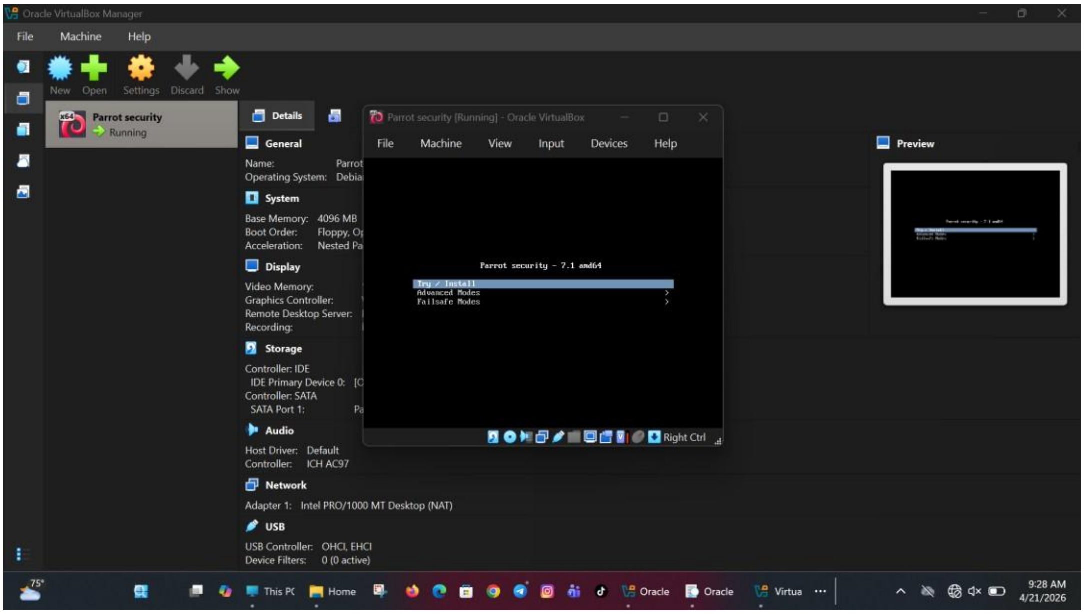

Figure 8: Starting virtual machine installation process.

Step 8: Select Language and Region

Choose your preferred language, country, and regional settings. This helps the system set the correct keyboard layout and location preferences.

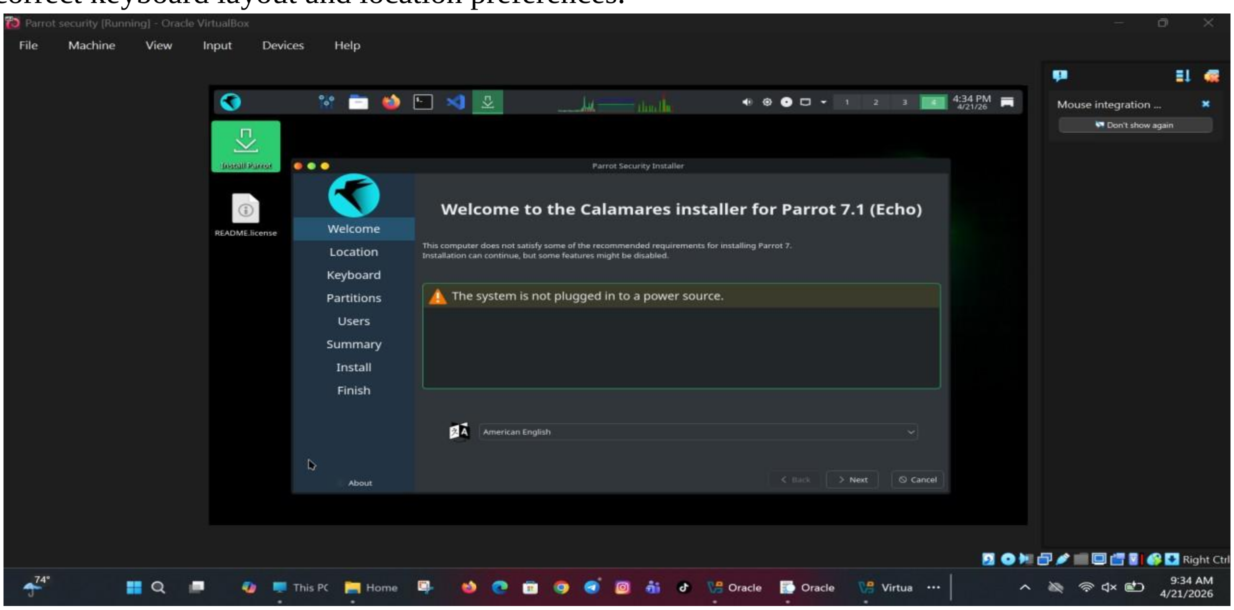

Figure 9: Selecting language and regional settings.

Step 9: Disk Partitioning

Select the automatic partitioning option for easier installation. This allows the system to create the required partitions without manual setup.

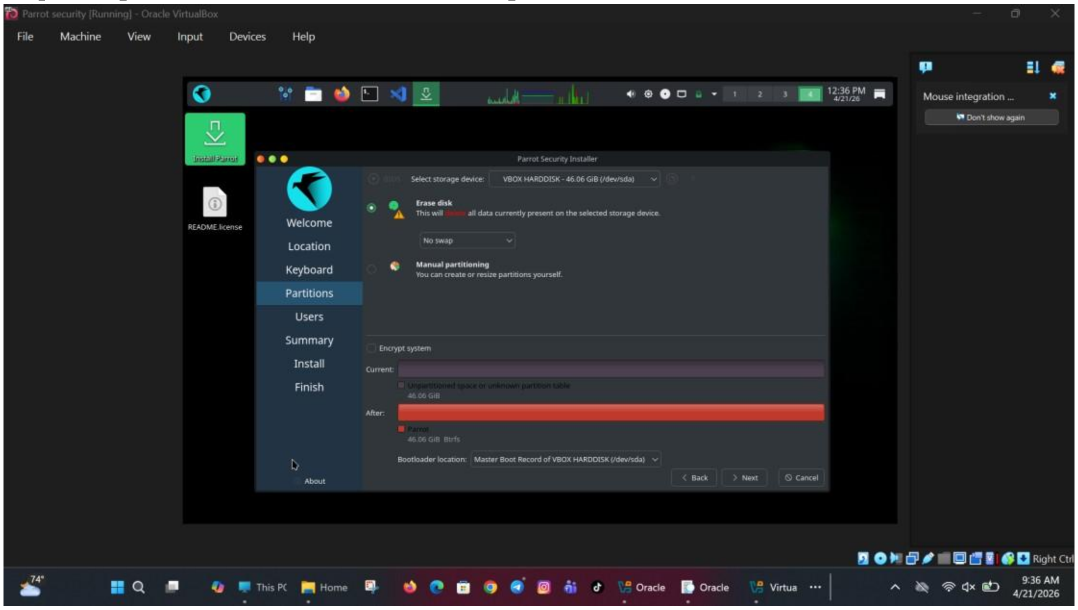

Figure 10: Choosing automatic disk partitioning option.

Step 10: Create User Account

Enter your full name, username, and password for the new system account. This account will be used later for logging into the operating system..

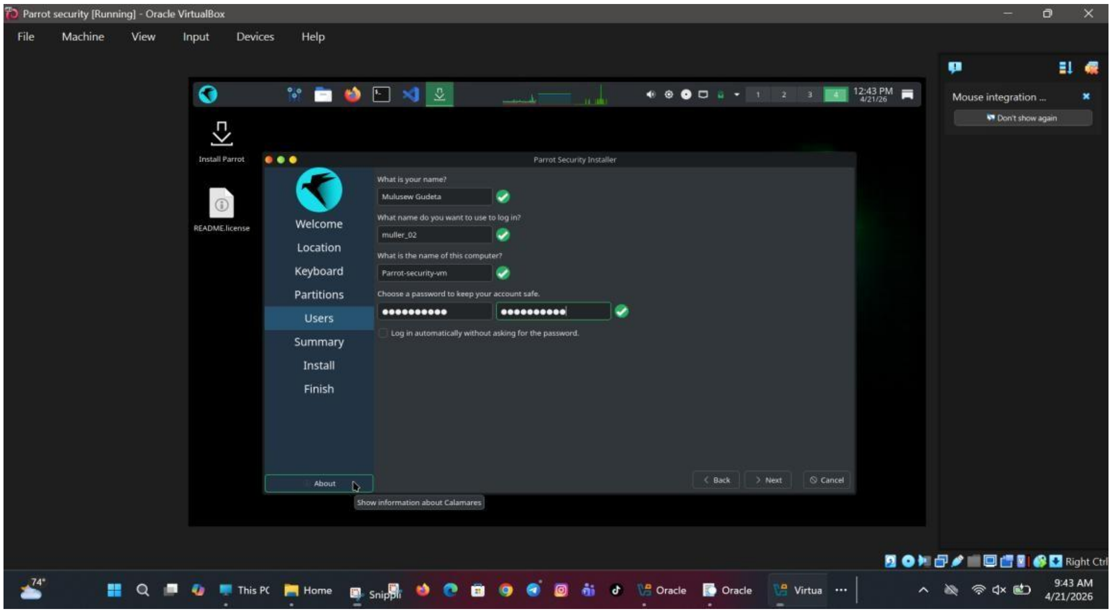

Figure 11: Creating username and password account

Step 11: System Installation

The system will now install all the necessary files automatically. This process may take a few minutes depending on your computer speed.

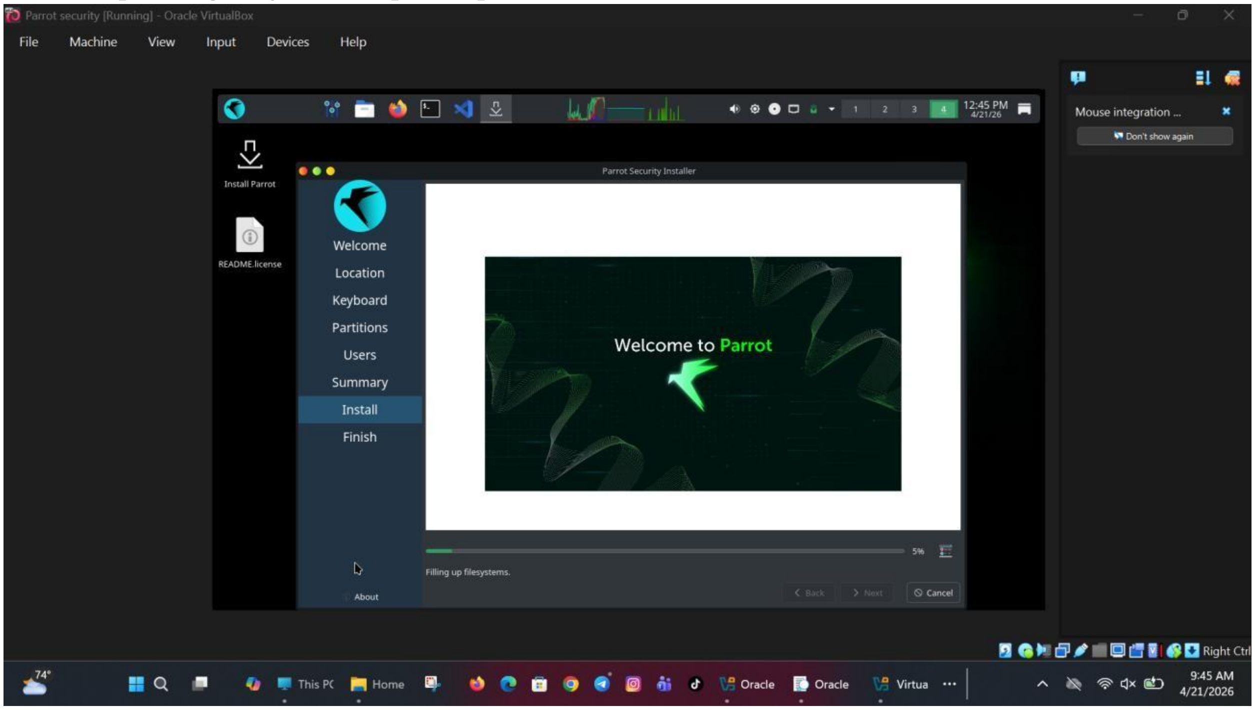

Figure 12: Installing Parrot Security OS files.

Step 12: Install Bootloader

Choose “Yes” to install the GRUB bootloader. This is important because it helps the system start properly after installation.

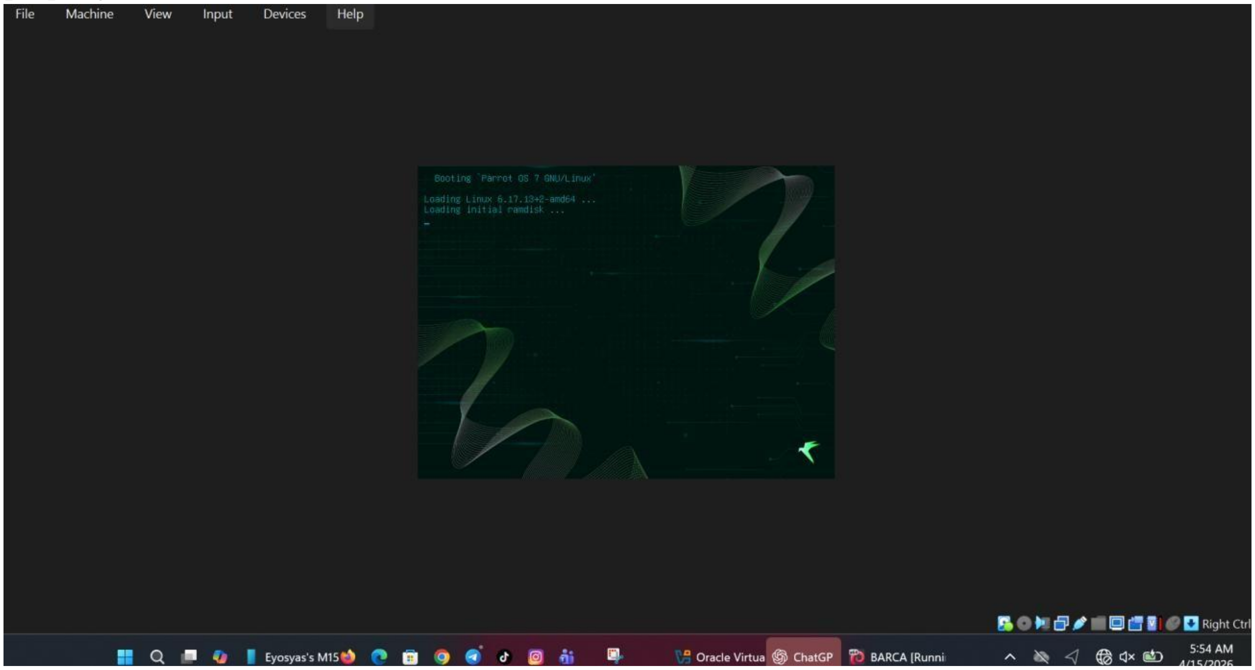

Figure 13: Confirming GRUB bootloader installation.

Step 13: Complete Installation

After the installation is finished, restart the virtual machine. This completes the setup and prepares the system for first use.

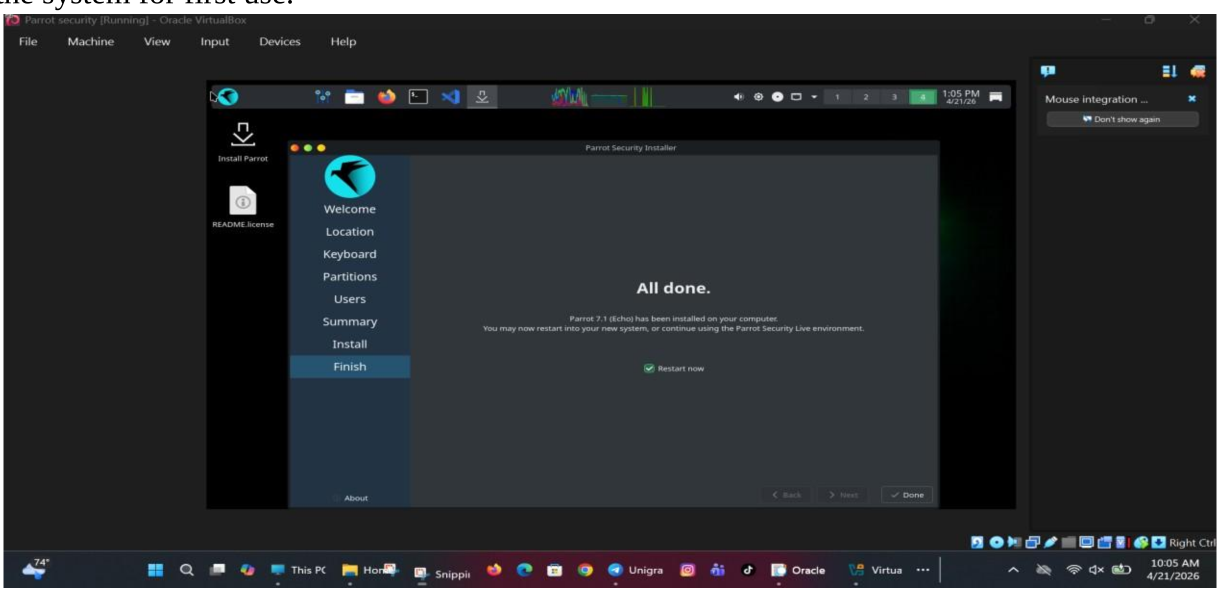

Figure 14: Completing installation and restarting system.

Step 14: Login and Desktop

Log in using your username and password created earlier. After login, the Parrot Security OS desktop environment will appear and the system will be ready to use.

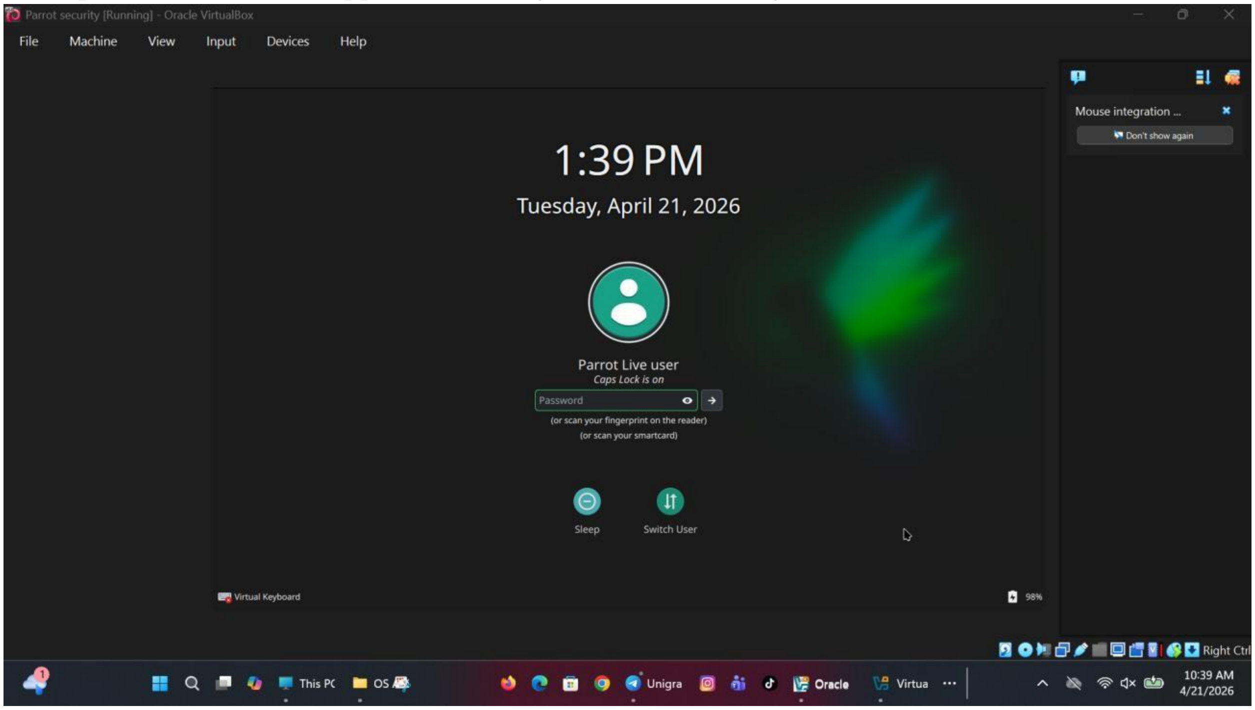

Figure 15: Parrot Security OS desktop during login.

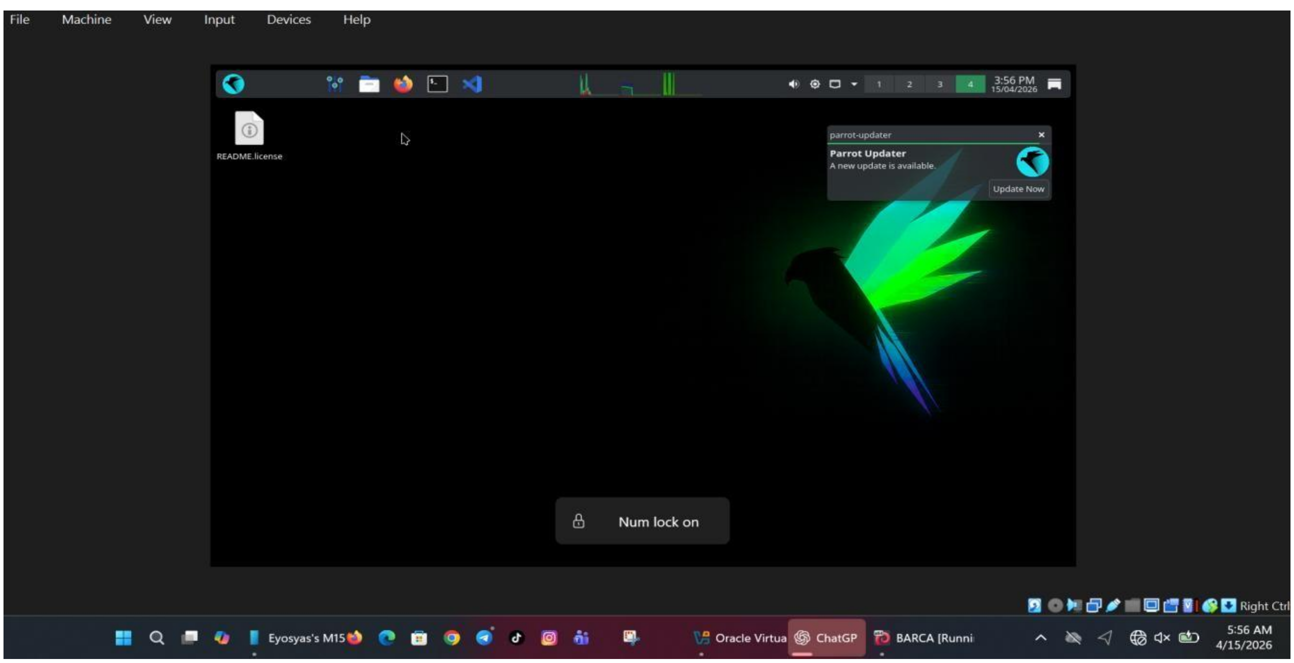

Figure 15: Parrot Security OS desktop after login.

44. Problems Encountered

During the installation of Parrot Security OS using Oracle VM VirtualBox, several technical challenges were faced. These problems affected the installation process and required troubleshooting before the system could work properly.

• **Virtual machine not starting properly** – After creating the virtual machine, it sometimes failed to boot or displayed startup errors. This usually happened because hardware virtualization was disabled in the BIOS settings.

• **Black screen after boot** – In some cases, the virtual machine started but only showed a black screen instead of the installation menu. This was mainly caused by incorrect display settings in VirtualBox.

• **Slow system performance** – The system became slow during installation and while opening applications because the allocated RAM and processor cores were too low for Parrot Security OS to run smoothly.

• **Network connectivity issues** – The guest operating system could not connect to the internet, which made downloading updates and accessing online resources difficult.

• **ISO file not recognized** – Sometimes the Parrot Security OS ISO file was not detected by VirtualBox, preventing the system from starting the installation process.

5. Solutions

Each problem was solved using the correct configuration and troubleshooting steps to ensure successful installation and better system performance.

• **Enabled virtualization in BIOS** – Hardware virtualization technology such as Intel VT-x or AMD-V was enabled from the BIOS settings of the host computer to allow the virtual machine to start properly.

• **Changed display settings to VMSVGA** – The graphics controller was changed to VMSVGA in VirtualBox settings, which solved the black screen problem and improved display compatibility.

• **Increased RAM and processor cores** – More RAM and CPU cores were assigned to the virtual machine to improve speed, stability, and overall system performance.

• **Set network mode to NAT** – The network adapter was configured to NAT (Network Address Translation), allowing the guest operating system to use the host computer's internet connection.

• **Reattached ISO file correctly** – The ISO installation file was removed and attached again from the storage settings in VirtualBox so that the system could boot properly from the installation media.

6. Filesystem Support

Parrot Security OS supports multiple file systems to provide better storage management, compatibility, and system performance depending on user needs.

• **ext4** – This is the default file system used in Parrot Security OS because of its speed, reliability, strong security, and journaling capability. It helps protect data during unexpected shutdowns or system crashes.

• **Btrfs** – This file system offers advanced features such as snapshots, backups, and improved storage management. It is useful for users who need better system recovery and data protection.

• **FAT32 / exFAT** – These file systems are mainly used for compatibility with external storage devices such as USB flash drives, memory cards, and portable hard drives.

• **NTFS** – This file system allows Parrot Security OS to interact with Windows-based storage systems, making it easier to access files stored on Windows partitions and transfer data between Linux and Windows.

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✅ Why ext4 is Preferred

- High performance
- Data integrity through journaling
- Wide compatibility in Linux environments

7. Advantages and Disadvantages

✅ Advantages

- Pre-installed security tools
- Lightweight and fast performance
- Strong privacy and anonymity features
- Open-source and regularly updated

❌ Disadvantages

- Not beginner-friendly
- Requires Linux knowledge
- High system resource usage for advanced tools
- Limited support for certain hardware

8. System Call: shutdown()

The `shutdown()` system call is used in network programming to control socket communication.

It allows a process to disable sending, receiving, or both operations on a socket connection. This is particularly useful when a program wants to close communication in a controlled manner without terminating the entire connection abruptly. **Syntax**

```
int shutdown(int sockfd, int how);
```

Modes

- SHUT_RD → Stops receiving data
- SHUT_WR → Stops sending data
- SHUT_RDWR → Stops both

Importance

- Prevents data corruption
- Ensures proper connection termination
- Improves system stability

Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is the process of creating a virtual version of a physical computer system. It allows one physical computer to run multiple operating systems at the same time by creating virtual machines (VMs). Each virtual machine behaves like a real computer with its own operating system, memory, storage, processor, and network settings. This means a user can use Windows as the main operating system and still run Linux, Ubuntu, or Parrot Security OS inside a virtual machine without needing another physical computer. Software like Oracle VM VirtualBox is commonly used to create and manage these virtual machines.

Why Virtualization?

Virtualization is important because it provides many benefits in modern computing. One major reason is that it creates a safe testing environment where users can install software, test applications, and practice system administration without affecting the main operating system. If a problem occurs inside the virtual machine, the host system remains safe. It also helps reduce costs because one physical computer can run several virtual machines, which reduces the need for multiple devices, electricity, and maintenance expenses. Another advantage is efficient resource usage, as CPU power, RAM, and storage can be shared among different virtual machines based on user needs. In addition, virtualization provides system isolation, meaning if one virtual machine crashes or becomes infected, the others continue working normally without being affected.

How it Works

Virtualization works through special software called a hypervisor. A hypervisor creates virtual hardware such as virtual CPU, RAM, hard disk, and network adapters, allowing multiple operating systems to run independently on one physical machine. It manages the communication between the physical hardware and the virtual machines while ensuring that each system remains separate and secure. There are two main types of hypervisors: Type 1, which runs directly on hardware, and Type 2, which runs on top of an existing operating system. Oracle VM VirtualBox is a Type 2 hypervisor because it runs on systems like Windows and allows users to install guest operating systems such as Parrot Security OS. This makes virtualization a powerful and essential technology in modern operating systems for education, business, and cybersecurity training.

Conclusion

This project provided a valuable hands-on learning experience in operating systems and system programming. Instead of only learning theory, it allowed practical work by installing and using Parrot Security OS in a virtual environment.

Through this process, I gained a better understanding of how operating systems function, especially the importance of proper configuration and attention to detail. Virtualization proved to be very useful, as it created a safe space to test and experiment without affecting the main system.

Working with Parrot Security OS also highlighted the strength of Linux-based systems in cybersecurity, while showing that they require some technical knowledge to use effectively. Learning about the `shutdown()` system call further deepened my understanding of how systems manage network communication.

Overall, this project helped me connect theory with practice and build a strong foundation for future studies in system administration, networking, and cybersecurity.

Future Outlook / Recommendation

For future improvement, it is recommended to explore more advanced tools available in Parrot Security OS, especially those used for penetration testing and security analysis. Practicing ethical hacking in controlled lab environments will help build real-world skills in a safe and legal way.

Learning scripting languages such as Bash and Python is also important, as they can be used to automate tasks and improve efficiency when working with Linux systems. In addition, gaining deeper knowledge of Linux system administration—such as managing users, processes, and networks—will strengthen overall technical ability.

It is also helpful to experiment with multiple virtual machines to better understand networking and system interaction. Finally, staying updated with the latest trends and developments in cybersecurity will ensure continuous learning and improvement.

Overall, consistent practice and curiosity will play a key role in building strong skills for future academic and professional success.

References

1. Oracle. *VirtualBox Official Documentation*.
2. Parrot Security Team. *Parrot Security OS Official Documentation*.
3. Linux Manual Pages. *shutdown() System Call Guide*.
4. Debian Documentation Project. *Linux Filesystem Documentation*.